Project Synopsis

Project Title

AI-Powered Movie Recommendation System

Problem Statement

Users often struggle to find movies that match their interests because of the huge amount of content available on streaming platforms. Traditional search methods are time-consuming and may not provide personalized recommendations.

Objective of the Project

To develop an AI-based movie recommendation system that suggests relevant movies based on user preferences and movie similarities, improving the user experience.

Proposed Solution / Description

The system will use machine learning techniques to analyze movie data such as genres, cast, keywords, and ratings. Based on user input, it will recommend similar movies through an interactive web interface.

Tools & Technologies to be Used

Python, Pandas, NumPy, Scikit-learn, Streamlit, TMDB Dataset, Pickle, VS Code/Jupyter Notebook, Git (optional).

Methodology / Approach

1. Collect and preprocess the movie dataset.
2. Clean and merge relevant features.
3. Convert text features into vectors using CountVectorizer/TF-IDF.
4. Calculate movie similarity using Cosine Similarity.
5. Build a recommendation engine.
6. Develop a Streamlit interface for user interaction.
7. Test and evaluate the recommendations.

Expected Outcome

A user-friendly application capable of providing fast and accurate personalized movie recommendations, helping users discover movies based on their interests.

Application / Future Scope

The system can be integrated with OTT platforms, enhanced with deep learning and collaborative filtering, support multiple languages, provide real-time recommendations, and recommend TV shows or web series.